College of Engineering – Department of Electronics Engineering

BACHELOR OF SCIENCE IN ELECTRONICS ENGINEERING
Academic Year 2025-2026
CMO No. 101 s. 2017, CMO No. 20 s. 2013, CMO No. 39 s. 2021, CMO No. 40 s. 2021

Curriculum Description

Electronics Engineering is the branch of engineering that integrates available and emerging technologies with knowledge of mathematics, natural, social and applied sciences to conceptualize, design, and implement new, improved, or innovative electronic, computer and communication systems, devices, goods, services and processes.

The program also aims to develop graduates with ethical behavior, integrity, social responsibility, effective communication, teamwork, and leadership skills, while promoting lifelong learning to keep up with the continuing technological advancement.

Program Education Objectives

The electronics engineering alumni three to five years after graduation shall:
PEO 1: Practice as a specialist in solving complex engineering problems driving improvements and innovations, while taking into consideration the environmental, social, and economical requirements, etc.
PEO 2: Assume leadership/influential position in industry, academe, government, or private sector within consideration to social and ethical responsibility.
PEO 3: Engage in lifelong learning through the following, but not limited to, further studies, research, certifications, promotions, and other personal and professional development activities.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:
SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex electronics engineering problems;
SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;
SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.
SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
SO5 Problem Analysis. Identify, formulate, and solve complex electronics engineering problems by applying principles of engineering, science, and mathematics;
SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.
SO7 Communication. Communicate effectively on complex electronics engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

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SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex electronics engineering activities;

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
Differential Calculus	4	0	4
Integral Calculus	4	0	4
Probability and Statistics for AI	3	0	3
Differential Equations	3	0	3
Discrete Mathematics	3	0	3
Subtotal	**17**	**0**	**17**
B. Natural/ Physical Sciences			
Chemistry for Engineers	3	3	4
Engineering Physics 1	3	3	4
Engineering Physics 2	3	3	4
Subtotal	**9**	**9**	**12**
C. Basic Engineering Sciences			
Introduction to Engineering	0	3	1
Engineering Drawing	0	3	1
Computer-Aided Design	0	0	1
Engineering Economics	3	0	3
Engineering Management	2	0	2
Technopreneurship	3	0	3
Subtotal	**8**	**6**	**11**
D. Allied Courses			
Materials Science and Engineering	3	0	3
Computer Programming for ECE	0	6	2
Circuits 1	3	3	4
Circuits 2	3	3	4
Environmental Science and Engineering	3	0	3
Basic Occupational Safety & Health	3	0	3
Subtotal	**15**	**12**	**19**
E. Professional Courses			
Electromagnetics	4	0	4

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ECE Laws and Professional Ethics	3	0	3
Programming for Electronic Tests and Designs	0	3	1
Advanced Engineering Mathematics for ECE	3	3	4
Electronic Devices and Circuits	3	3	4
Digital Principles and Logic Design	3	3	4
Electronic Circuit Analysis and Design	3	3	4
Principles of Communication Systems	3	3	4
Feedback and Control Systems	3	3	4
Numerical Methods & Analysis for ECE	2	3	3
Electronic Systems and Design 1	2	3	3
Electronic Systems and Design 2	1	3	3
Data Communications	3	3	4
Transmission Media and Antenna Systems	3	3	4
Signals, Spectra, and Signal Processing	3	3	4
Modulation and Coding Techniques	3	3	4
Microprocessor and Microcontroller Systems and Design	3	3	4
Methods of Research	3	0	3
ECE Capstone Design 1	0	3	1
ECE Capstone Design 2	0	3	1
ECE Seminars/Colloquium	0	3	1
ECE Review with Comprehensive Examination	0	6	2
Subtotal	**48**	**60**	**68**
F. Elective Courses			
ECE Elective 1	2	3	3
ECE Elective 2	2	3	3
ECE Elective 3	2	3	3
Subtotal	**6**	**9**	**9**
G. On-the-Job Training			
Internship/OJT (320 hrs)			**4**
TOTAL TECHNICAL COURSES	**103**	**96**	**140**
II. NON-TECHNICAL COURSES			
A. General Education Courses			
Understanding the Self	3	0	3
Mathematics in the Modern World	3	0	3
The Contemporary World	3	0	3
Readings in Philippine History	3	0	3
Purposive Communication	3	0	3
Ethics	3	0	3
Art Appreciation	3	0	3
Science, Technology and Society	3	0	3
Subtotal	**24**	**0**	**24**
B. General Education Electives			
Linear Algebra for AI	3	0	3
Machine Learning and Neural Networks	2	3	3
The Entrepreneurial Mind	3	0	3
Life and Works of Rizal	3	0	3
Subtotal	**11**	**3**	**12**
C. Physical Activities Toward Health and Fitness (PATHFit)			

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Movement Competency Training	2	0	2
Exercise-based Fitness Activities	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2	2	0	2
Subtotal	**8**	**0**	**8**
National Service Training Program 1	3	0	3
National Service Training Program 2	3	0	3
Subtotal	**6**	**0**	**6**
TOTAL NON-TECHNICAL COURSES	**49**	**3**	**50**
GRAND TOTAL	**152**	**99**	**190**

SUMMARY

Courses	Number of Units
I. TECHNICAL COURSES	
A. Mathematics	17
B. Natural/Physical Sciences	12
C. Basic Engineering Sciences	11
D. Allied Courses	19
E. Professional Engineering Courses	68
F. Elective Courses	9
G. On-the-Job Training	4
II. NON-TECHNICAL COURSES	
A. General Education Courses	24
B. General Education Elective and Mandated Courses	12
C. Physical Activities Toward Health and Fitness (PATHFit)	8
D. National Service Training Program	6
GRAND TOTAL	**190**

CURRICULUM STRUCTURE

FIRST YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
GEd 101	Understanding the Self	3	3	0		
GEd 102	Mathematics in the Modern World	3	3	0		
GEd 106	Purposive Communication	3	3	0		
CHEM 101	Chemistry for Engineers	4	3	3		
MATH 101	Differential Calculus	4	4	0		
ENGG 101	Introduction to Engineering	1	0	3		
ENGG 102	Engineering Drawing	1	0	3		
PATHFit 1	Movement Competency Training	2	2	0		
NSTP 111	National Service Training Program 1	3	3	0		
	TOTAL	**24**	**21**	**9**		

FIRST YEAR

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SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
GEd 104	The Contemporary World	3	3	0		
GEd 105	Readings in Philippine History	3	3	0		
GEd 108	Art Appreciation	3	3	0		
GEd 109	Science, Technology and Society	3	3	0		
CPE 104	Computer Programming for ECE	2	0	6		
MATH 102	Integral Calculus	4	4	0	MATH 101	
PHYS 104	Engineering Physics 1	4	3	3	MATH 101	MATH 102
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1	
NSTP 121	National Service Training Program 2	3	3	0	NSTP 111	
	TOTAL	**27**	**24**	**9**		

FIRST YEAR						
MIDTERM						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
GEdE 101	The Entrepreneurial Mind	3	3	0		
GEd 107	Ethics	3	3	0		
ENGG 118	Materials Science and Engineering	3	3	0	CHEM 101	
	TOTAL	**9**	**9**	**0**		

SECOND YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
MATH 103	Differential Equations	3	3	0	MATH 102	
PHYS 105	Engineering Physics 2	4	3	3	PHYS 104	
ENGG 103	Computer-Aided Design	1	0	3	ENGG 102	
EE 207	Circuits 1	4	3	3	MATH 102	PHYS 105
ECE 201	Electromagnetics	4	4	0		MATH 103, PHYS 105
CpE 202	Discrete Mathematics	3	3	0	MATH 102	
AI 101	Linear Algebra for AI	3	3	0		
ECE 202	ECE Laws and Professional Ethics	3	3	0		
PATHFit 3	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2	2	0	PATHFit 1 & 2	
	TOTAL	**27**	**24**	**9**		

SECOND YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
ECE 203	Programming for Electronic Tests and Designs	1	0	3	CPE 101	
EE 208	Circuits 2	4	3	3	EE 207	
ECE 204	Advanced Engineering Mathematics for ECE	4	3	3	MATH 103	
ECE 205	Electronic Devices and Circuits	4	3	3	EE 207, PHYS 105	
ECE 206	Digital Principles and Logic Design	4	3	3	EE 207	ECE 205

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PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, and Adventure Activities 2	2	2	0	PATHFit 1 & 2	
ENGG 113	Environmental Science and Engineering	3	3	0		
AI 102	Probability and Statistics for AI	3	3	0	AI 101	
	TOTAL	**25**	**20**	**15**		

THIRD YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
ECE 304	Numerical Methods & Analysis for ECE	3	2	3	ECE 204, CPE 101	
ECE 301	Electronic Circuit Analysis and Design	4	3	3	ECE 205	
ECE 302	Principles of Communication Systems	4	3	3		ECE 301
ECE 303	Feedback and Control Systems	4	3	3	ECE 204, EE 208	
ECE 305	Electronic Systems and Design 1	3	2	3	ECE 205	
ECEE 401	ECE Elective 1	3	2	3	3rd year standing	
AI 103	Machine Learning and Neural Networks	3	2	3	AI 102	
	Total	**24**	**17**	**21**		

THIRD YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
ECE 306	Electronic Systems and Design 2	2	1	3	ECE 305, ECE 206	
ECE 310	Modulation and Coding Techniques	4	3	3	ECE 302	
ECE 307	Data Communications	4	3	3	ECE 302	
ECE 308	Transmission Media and Antenna Systems	4	3	3	ECE 302, ECE 201	
ECE 309	Signals, Spectra, and Signal Processing	4	3	3	ECE 204	
ENGG 116	Methods of Research	3	3	0	3rd year standing	
ECEE 402	ECE Elective 2	3	2	3	ECEE 401	
	Total	**24**	**18**	**18**		

THIRD YEAR						
MIDTERM						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
ECE 311	Microprocessor and Microcontroller Systems and Design	4	3	3	CPE 101, ECE 206	
ENGG 114	Basic Occupational Safety and Health	3	3	0		
ENGG 104	Engineering Economics	3	3	0		
	Total	**10**	**9**	**3**		

FOURTH YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
ENGG 106	Engineering Management	2	2	0		
ECE 401	ECE Capstone Design 1	1	0	3	ENGG 116	
ENGG 105	Technopreneurship	3	3	0	4th year standing	

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ENGG 117	Internship/OJT (320 hrs)	4	320 hours		4th year standing	
	Total	**10**	**5**	**3**		

FOURTH YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**lab**	**Prerequisite(s)**	**Co-requisite(s)**
ECE 404	ECE Review with Comprehensive Examination	2	0	6	Graduating	
ECE 403	ECE Seminars/Colloquium	1	0	3	4th year standing	
ECE 402	ECE Capstone Design 2	1	0	3	Graduating	
ECEE 403	ECE Elective 3	3	2	3	ECEE 402	
GEd 103	Life and Works of Rizal	3	3	0		
	Total	**10**	**5**	**15**		